

Questions

1. How does polymorphism improve flexibility in object-oriented design?

Answer:

Polymorphism enhances flexibility in that dissimilar objects can be treated as similar objects and behave differently. This implies that there is only one method or interface to many types of objects, and the object can respond in a different manner. Consequently, you can add new classes or alter behavior without having to modify existing code, resulting in easier program extendibility and maintenance.

2. What is the advantage of using inheritance to extend functionality?

Answer:

Through inheritance, a new class is enabled to reuse and extend the attributes of an already existing class. This helps decrease the duplication of codes, saves time during development and make the code more structured. It also enables the developers to add or override the functionality without having to write the whole program.

Observations

These are the important concepts in OOP since they assist in increasing flexibility and organization of programs. Polymorphism enables one to have varying objects being treated alike whilst carrying out different operations and thus the code is more readily extended. The reuse of code is encouraged through inheritance making it possible to have new classes building on the old ones and hence repetition is minimized, and therefore maintenance is made easier.